

Drainage lab • evidence and method

Soil: let the water through

Name _____ Date _____

Constructed sample data. No practical is needed. Each trial uses fresh, equally prepared 100 cm³ samples; 60 mL added over 10 seconds; read at 2 minutes from pour start.

Soil	Trial 1: mL out	Trial 2: mL out
A	42	40
B	26	28
C	10	12

A • Plan

All pots: 60 mL water. Read all at 2 minutes.

B • Plan

A gets 60 mL. B gets 100 mL. Compare them.

C • Plan

All pots: 100 cm³ soil, loosely filled without tamping.

D • Plan

Pack C tightly. Leave A and B loose.

W1. Choose two useful cards. Repair the other two.

Route 1 • Compare and explain

Use the constructed sample data on the evidence page. An adult may read or write your spoken answer.

S1. Which trial 1 volume is greatest?

- A. 42 mL
- B. 26 mL
- C. 10 mL

S2. In this test, soil ___ let most water through because ___.

S3. For a fair comparison, which plan should we use?

- A. All samples get 60 mL and 2 minutes.
- B. Give each sample a different amount and time.

S4. Why might we do a second trial?

S5. Draw or describe how water can pass through soil.

Route 2 • Record a test

Choose constructed data or a teacher-approved practical. Label your data source.
Practical: follow the complete method on the evidence page with identical pots/filters and matched starting moisture.

T1. My data source is _____. The conditions I will keep the same are _____.

T2: Soil	Water in (mL)	Time (min)	Water out (mL)
A			
B			
C			

T3. Compare the three collected volumes. Which drained most in this test? Use numbers.

T4. The constructed second trial gives A 40, B 28, C 12 mL. Does it show the same order as the constructed first trial?

T5. Record a problem or one improvement. If there was no practical, suggest a possible problem.

Route 3 • Challenge the test

Use the constructed sample data on the evidence page.

X1. Use both trials to explain the drainage pattern.

A new test uses equal pre-weighed portions of the same soil, with matched starting moisture, pots and filters. P is loose; Q is tightly packed. Each receives 60 mL over 10 seconds and is read after 2 minutes from pour start. P collects 34 mL; Q collects 18 mL. These are constructed sample data.

X2. What changed in this new test? What does it suggest?

X3. Why would changing both soil type and packing make it harder to explain a result?

X4. A designer wants to choose soil for a growing bed. What else would you investigate?

Exit and reflection

Use the constructed sample data for E1. Respond by pointing, drawing, writing or speaking.

E1. Which sample let most water through? Include one volume.

E2. Name one condition that should be kept the same.

E3. Does this test choose the best soil for every plant? Explain.

R1. One useful improvement we heard was ____.

R2. Next time I would choose the role ____ because ____.
